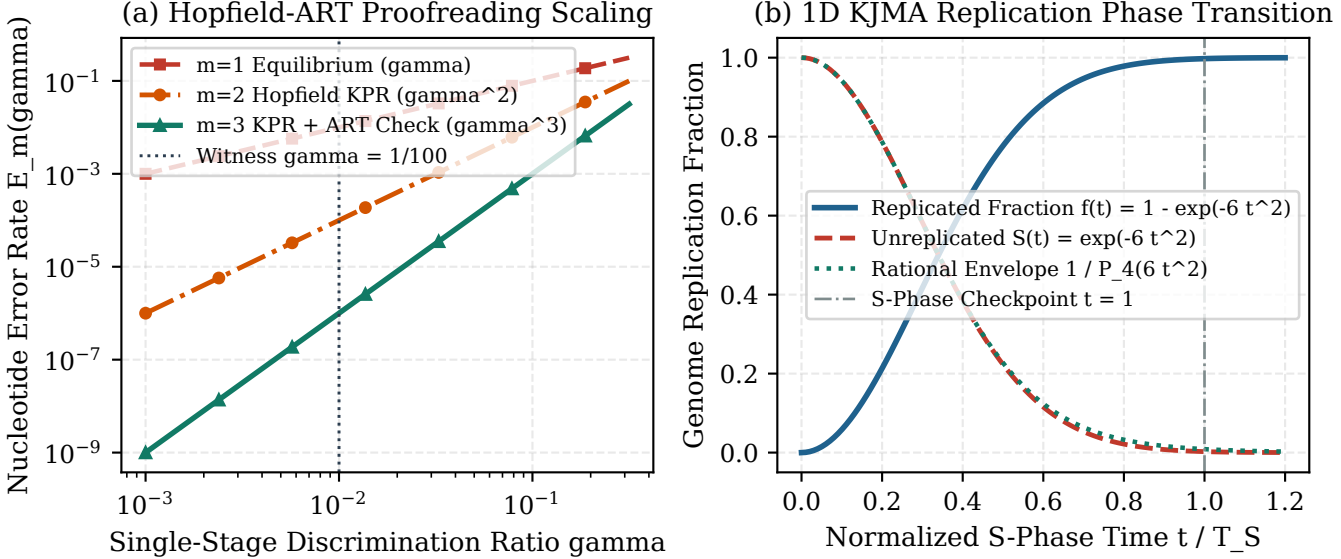
(a) Hopfield-ART Proofreading Scaling
m=1 Equilibrium (gamma)
m=2 Hopfield KPR (gamma^2)
m=3 KPR + ART Check (gamma^3)
Witness gamma = 1/100
Nucleotide Error Rate E_m(gamma)
Single-Stage Discrimination Ratio gamma
(b) 1D KJMA Replication Phase Transition
Replicated Fraction f(t) = 1 - exp(-6 t^2)
Unreplicated S(t) = exp(-6 t^2)
Rational Envelope 1 / P_4(6 t^2)
S-Phase Checkpoint t = 1
Genome Replication Fraction
Normalized S-Phase Time t / T_S